

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.
20										817	USE ONLY FOR TIE IN
20	U04		MODULAR DBB	K03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
30	U04		MODULAR DBB	K03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	BA8	NOT TO BE USED UNLESS REQUIRED BY SITE
60	U04	VAW	MODULAR DBB	M03EC02	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	NT5	TO BE USED IN NON-PIGGED LINES ONLY
70										SB6	USE JACKSCREWS FOR PADDLE
70							***** BALL VALVES *****				BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
80										UJ2	USE 1 1/2IN WITH THERMOWELL
80	U04	VB2	BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
90	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		
100	U04	VB2	BALL	A03AC07	6	8	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
110	U04		BALL	A03AC08	6	8	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
120	U04	VB2	BALL	B03AC01	0.5	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
130	U04	VB2	BALL	B03AC01	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
140	U04		BALL	B03AC02	0.5	2	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
150	U04	VA5	BALL	B03AC02	3	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
160	U04	VB2	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		
170	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0		
180	U04	VB2	BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
190	U04	VA5	BALL	B03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
200	U04	VB2	BALL	B03JC07	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0		
210	U04	VA5	BALL	B03JC08	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0		
220											
220							***** RISING STEM BALL VALVES *****				
230	U04	VA5	BALL	R03AC06	2	6	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
240											
240							***** BUTTERFLY VALVES *****				
250			BUTTERFLY	T03AC61	24	24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
260			BUTTERFLY	T03JC61	28	28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
270											
270							***** GATE VALVES *****				
280											
280			GATE	W03AC70	3	6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
290			GATE	W03AC71	24	24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
300			GATE	W03JC71	28	28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A	0	0		
310											
310							***** GLOBE VALVES *****				
320	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
330			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
340											
340							***** CHECK VALVES *****				
350	U04		CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
360			CHECK	C03AC45	2	6	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
370			CHECK	C03AC52	24	24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16	0	0		
380			CHECK	C03JC52	28	28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0		
390											
390							***** PIPE ABOVE GROUND *****				
400											
400			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
410			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
420			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
430											
430			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
440			PIPE	9001254	6	16	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0		
450			PIPE	9001256	18	20	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE	0	0		
460			PIPE	9001311	24	24	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
470			PIPE	9001493	28	28	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0		
480											
480		810	PIPE	5358512	0.5	0.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
490											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1.5 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO		MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C		12/16/2024	BY OZAC	U04	Page 1 of 4	300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
490							***** PIPE BELOW GROUND *****				
500											
510			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
520			PIPE	9001254CTD	6	16	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
530			PIPE	9001256CTD	18	20	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED	0	0		
540			PIPE	9001311CTD	24	24	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0		
550			PIPE	9001493CTD	28	28	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
560							***** FITTINGS ABOVE & BELOW GROUND*****				
560			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
570			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
580			ELBOW	9006566	4	4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60	0	0		
590	U04		ELBOW	9002190	6	8	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0		
600			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
610			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
620	U04		ELBOW	9006573	4	4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60	0	0		
630	U04		ELBOW	9006571	6	8	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0		
640	U04	BA8		9002156	6	6	ELL BW 90 DEG SR WT 6.35 MM A860 WPHY 60	0	0		
650											
650			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
660			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
670		NT5	TEE	9301610	4	4	TEE BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
680		NT5	TEE	9301609	6	6	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0		
690			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2		
700			TEE RED	5392454	0.75	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
710		NT5	TEE RED	9006664	2	2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4	4		
720		NT5	TEE RED	9006659	3	3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4	4		
730		NT5	TEE RED	9006658	3	3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6	6		
740		NT5	TEE RED	9006605	4	4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6	6		
750											
750			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
760			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
770			CAP	9005062	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
780			CAP	9005055	6	16	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0		
790			CAP	9005057	18	20	CAP BW WT 7.14 MM LTCS A860 WPHY-60	0	0		
800			CAP	9005058	24	24	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0		
810			CAP	9005060	28	28	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0		
820											
820			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2		
830			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3		
840			RED CONC	9006918	1.5	3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
850			RED CONC	9006915	3	3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6	6		
860	U04		RED CONC	9006926	3	3	REDUCER CONC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
870	U04		RED CONC	9006930	4	4	REDUCER CONC WT 6.35 MM / 6.02 WT LTCS A860 WPHY-60	6	6		
880			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2		
890			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3		
900			RED ECC	9001372	1.5	3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
910			RED ECC	9002510	3	3	REDUCER ECC WT 6.35 MM/STD LTCS A420 WPL6	6	6		
920			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	10		
930			RED ECC	9002266	6	14	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	16		
940	U04		RED ECC	9002281	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
950	U04		RED ECC	9006103	3	3	REDUCER ECC WT 6.35 MM / STD LTCS A860 WPHY-60	6	6		
960			RED ECC	9002269	10	16	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	18	18		
970			RED ECC	9006501	12	16	REDUCER ECC WT 7.14 MM / 6.35MM LTCS A860 WPHY-60-W	20	20		
980			RED ECC	9006502	18	18	REDUCER ECC WT 7.14 MM LTCS A860 WPHY-60-W	20	20		
990			RED ECC	9004138	16	16	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60-W	24	24		
1000			RED ECC	9004139	18	20	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60-W	24	24		
1010			RED ECC	9004134	18	20	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	28	28		
1020			RED ECC	9004135	24	24	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	28	28		
1030							*****INDUCTION BENDS ABOVE GROUND*****				
1040											
1040			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1050			INDUC BEND	9001813	6	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1060			INDUC BEND	9001814	18	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1070			INDUC BEND	9001815	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1080			INDUC BEND	9001817	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1090			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1100			INDUC BEND	9001823	6	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1110			INDUC BEND	9001824	18	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		
1120			INDUC BEND	9001825	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1130			INDUC BEND	9001827	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1140			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1150			INDUC BEND	9001833	6	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1160			INDUC BEND	9001834	18	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1170			INDUC BEND	9001835	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1180			INDUC BEND	9001837	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1190										
1190							*****INDUCTION BENDS BELOW GROUND*****			
1200										
1200			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1210			INDUC BEND	9001813CTD	6	16	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1220			INDUC BEND	9001814CTD	18	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1230			INDUC BEND	9001815CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1240			INDUC BEND	9001817CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1250			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1260			INDUC BEND	9001823CTD	6	16	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1270			INDUC BEND	9001824CTD	18	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1280			INDUC BEND	9001825CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1290			INDUC BEND	9001827CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1300			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1310			INDUC BEND	9001833CTD	6	16	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1320			INDUC BEND	9001834CTD	18	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1330			INDUC BEND	9001835CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1340			INDUC BEND	9001837CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1350	U04		INDUC BEND	9007001CTD	6	6	INDUCTION BEND 45DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1360	U04		INDUC BEND	9007016CTD	6	6	INDUCTION BEND 45DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1370	U04		INDUC BEND	9007031CTD	6	6	INDUCTION BEND 90 DEG X 10D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1380	U04		INDUC BEND	9007043CTD	6	6	INDUCTION BEND 90 DEG X 15D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1390										
1390							***** OLETS ABOVE & BELOW GROUND*****			
1400										
1400	U04		OLETS	9006219	0.75	1.5	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4	
1410			OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	6	
1420			OLETS	9006408	4	4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	12	16	
1430	U04		OLETS	5505179	0.75	1	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	3	
1440			OLETS	9006411	4	4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	18	20	
1450			OLETS	9006701	6	6	WELDOLET WT 7.14 MM/6.35 MM LTCS A694 F60	18	20	
1460			OLETS	9006414	4	4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	24	24	
1470			OLETS	9004124	6	6	WELDOLET WT 7.92 MM/6.35 MM LTCS A694 F60	24	24	
1480			OLETS	9006427	4	4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	28	28	
1490			OLETS	9006424	6	10	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	28	28	
1500	U04		OLETS	5504380	1	1	WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1	0	0	
1510										
1510	U04		OLETS	5504020	0.75	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2	
1520	U04		OLETS	5504021	0.75	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3	
1530			OLETS	9002639	0.75	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4	
1540			OLETS	9002643	0.75	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	16	
1550	U04	UJ2	OLETS	9002651	1.5	1.5	NIPOFLANGE 300# RF 6.35 MM/160 WT LTCS A694 F60	6	6	
1560			OLETS	9002644	0.75	2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	18	20	
1570			OLETS	9002645	0.75	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	24	24	
1580			OLETS	9002647	0.75	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	28	28	
1590										
1590							***** FLANGES ABOVE & BELOW GROUND*****			
1600										
1600	U04		FLANGE	5556476	0.5	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1610	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1620			FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0	
1630			FLANGE	9001483	6	16	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0	
1640			FLANGE	9001484	18	20	FLG WN 300# RF LTCS A694 F60 WT 7.14MM	0	0	
1650			FLANGE	9001455	24	24	FLG WN 300# RF LTCS A694 F60 WT 7.92MM	0	0	
1660			FLANGE	9001461	28	28	FLG WN 300# RF LTCS A694 F60 WT 9.53 MM B16.47 SERIES A	0	0	
1670	U04		BLIND	5615576	0.5	3	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
1680			BLIND	9001450	4	24	FLG BLIND 300# RF LTCS A694 F60	0	0	
1690			BLIND	9001451	28	28	FLG BLIND 300# RF LTCS A694 F60 SERIES A	0	0	
1700	U04		TEE RED	5599703	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	300H5P

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1710	U04		FLANGE	9006752	3	3	FLG WN REDUCING 300# RF LTCS A694 F60 BORED TO SUIT WT 6.02MM	4	4	
1720	U04		FLANGE	5557036	1.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1730	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
1740			FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
1750			FLANGE	9001471	6	6	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0	
1760			FLANGE	9005389	4	4	FLG WN 900# RF LTCS A694 F60 WT 6.02MM	0	0	
1770	U04		FLANGE	5557393	2	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1780	U04	817	FLANGE	5558033	3	3	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH STD	0	0	
1790	U04	817	FLANGE	5558036	2	2	FLG WN 600# RTJ LTCS A350 LF2 CL.1 SCH XS	0	0	
1800	U04		FLANGE	9006755	6	6	FLG WN 900# RF LTCS A694 F60 WT 6.35MM	0	0	
1810										
1810	U04		FLG ADAPTER	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
1820										
1820							***** ORIFICE FLANGES *****			
1830										
1830	U04		ORIFICE FLG	9001559	4	4	WN ORIF SET 300# RF A694 F60 WT 6.02MM	0	0	
1840	U04		ORIFICE FLG	9001558	6	6	WN ORIF SET 300# RF A694 F60 WT 6.35MM	0	0	
1850										
1850							***** GASKETS *****			
1860										
1860			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1870			GASKET	5674032	28	28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1880	U04		GASKET	5673103	1.5	6	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1890	U04		GASKET	5673117	4	6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1900	U04		GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1910	U04	817	GASKET	5673202	2	3	GASKET 600# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1920										
1920		B1					***** BOLTING *****			
1930										
1930	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1940										
1940		SB6	BOLTING	5675675	12	28	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1950										
1950			SPEC BLIND	5749581	0.75	6	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1960										
1960			PADDLE	5750100	24	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1970			PADDLE	5750028	28	28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
1980			PADDLE	5750200	24	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1990			PADDLE	5750198	28	28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
2000										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1.5 MM	SOUR SERVICE - NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 80 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 80 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	300H5P